Complete Self-Similar Solution of the Subsonic Marshak Wave

Theory and Numerical Validation Structure

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Motivation

Inertial Confinement Fusion (ICF)

- **Hohlraum Modeling:** Crucial for radiation temperature prediction and experimental measurement accuracy[cite: 5, 6].
- **Energy Delivery:** Key to analyzing laser–walls coupling and refining pulse shaping techniques[cite: 7, 8].
- **Historical Context:** Builds upon classical foundations established by *Lindl (1995)*, *Kauffman (1995)*, and *Rosen (2005)*[cite: 9].

Equation of State (EOS) Verification

- Critical for laboratory astrophysics and high-energy density physics (HEDP) experiments under dynamic, time-varying pressure profiles *[Thomas 2006]*[cite: 10].

The Subsonic Wave Mechanism

- We treat a semi-infinite wall subjected to high-temperature boundary heating[cite: 43].
- Dynamic equilibrium is mediated strictly via **photon transport**[cite: 44].
- In the optically thick limit, an exceptionally sharp temperature front develops[cite: 45].

Subsonic Regime Conditioning

If the localized heat front travels **slower than the material speed of sound**, hydrodynamics decouples into two distinct zones: a **shock region** driven forward ahead of the high-temperature **ablation region**[cite: 46].

Our Work – Overview & Strategy

- **Decoupled Patching:** Instead of a single complex hydro-transport fit, we solve the ablation zone and the shock zone independently, later matching them via self-consistent boundary logic[cite: 69, 471].
- **Self-Similarity:** In each localized frame, we systematically neglect one governing dimensional parameter to yield exact self-similar ODEs[cite: 70].
- **Inter-Zone Coupling:** The thermodynamic properties calculated at the heat front serve directly as the driving boundary condition for the downstream shock equations[cite: 71].

Parameter Pool	**Ablation Region**	**Shock Region**
Density Profile	Assumed $\rho \to 0$ at source [cite: 95, 98]	Captures backgrou
Radiation Flux (B)	Fully Coupled [cite: 90, 94]	Neglected ($B \approx 0$)

Table: Strategic scaling assumptions across regions.

Statement of the Problem

The 1D radiation-hydrodynamic transport framework in Lagrangian mass coordinates m is stated as[cite: 86]:

$$\frac{\partial V}{\partial t} = \frac{\partial u}{\partial m}$$

$$\frac{\partial u}{\partial t} = -\frac{\partial P}{\partial m}$$

$$\frac{\partial e}{\partial t} + P\frac{\partial V}{\partial t} = \frac{\partial}{\partial m}\left(\frac{c}{3\kappa_R}\frac{\partial (aT^4)}{\partial m}\right)$$

Combined Nonlinear Energy Equation

By tracking power-law material properties, the energy diffusion scales as[cite: 90]:

$$\frac{1}{r}\frac{\partial PV}{\partial t} + P\frac{\partial V}{\partial t} = B\frac{\partial}{\partial m}\left(V^{\lambda}\frac{\partial}{\partial m}\left(PV^{1-\mu}\right)^{\frac{4+\alpha}{\beta}}\right)$$

where the structural constant is $B = \frac{16\sigma}{3(4+\alpha)}g(rf)^{\frac{-(4+\alpha)}{\beta}}$[cite: 90].

Thermodynamic & Material Modeling

To construct a closed analytic framework, variables are mapped using power-law dependencies[cite: 73]:

Boundary Heating Profile

$$T(t) = T_0 t^{\tau}$$

Specific Internal Energy

$$e = fT^{\beta}\rho^{-\mu}$$

Rosseland Mean Opacity

$$\frac{1}{\kappa_R} = gT^{\alpha}\rho^{-\lambda}$$

Equation of State (EOS)

$$P = r\rho e \equiv (\gamma - 1)\rho e$$

Background density is initialized uniformly to ρ_0 ahead of the wavefront[cite: 78].

The Ablation Region Properties

The ablation zone is completely characterized by four dimensional parameters: $B, T_0, \rho_0(V_0)$, and t[cite: 94]. Transforming to the dimensionless similarity coordinate ξ scales the structural variables[cite: 101]:

- **Lagrangian Coordinate Scaling:** [cite: 101]

$$m = \xi \cdot \left(B^{2-\mu} T_0^{8+2\alpha-2\beta+\beta\lambda-(4+\alpha)\mu} t^{2+2(4+\alpha-\beta)\tau-\mu(3+(4+\alpha)\tau)+\lambda(2+\beta\tau)} \right)^{\frac{1}{4+2\lambda-4\mu}}$$

- **Local Structural Functions:** Evaluated using dimensional scaling tracks [cite: 102]

$$X(m,t) = \tilde{X}(\xi) B^{w_{X1}} T_0^{w_{X2}} t^{w_{X3}}$$

Boundary Conditions for Ordinary Differential Equations

Boundary bounds are enforced exactly at the zero-volume ablation edge ξ_F[cite: 106]:

$$\tilde{V}(\xi_F) = 0, \quad \left.\frac{\partial \tilde{V}}{\partial \xi}\right|_{\xi_F} = 0, \quad \tilde{P}(0) = 0, \quad \tilde{u}(\xi_F) = 0$$

The Shock Region Integration

Once the ablation scale maps the pressure at the backend boundary, the shock zone is localized via $P(t) = P_0 t^{\tau_s}$[cite: 139]. Neglecting active radiation transport $(B \to 0)$ yields classic strong shock relations[cite: 140, 141]:

Rankine-Hugoniot Mass, Momentum, and Energy Conditions

$$\frac{D - u_S}{V_S} = \frac{D}{V_0}, \quad P_S = \frac{D u_S}{V_0}, \quad P_S u_S V_0 = D\left(e_S + \frac{u_S^2}{2}\right)$$

- **EOS Divergence:** To retain generality, the material model inside the shock domain is permitted to utilize alternative polytropic definitions $(r \neq r')$[cite: 144, 146].
- **Shock Coordinate Mapping:** mass variables scale via[cite: 180]:

$$m_S = \xi P_0^{1/2} V_0^{-1/2} t^{1+\frac{\tau_S}{2}}$$

Simulation Setup: Case 1 ($\tau = 0$)

Medium Framework: Gold Target

The numerical validation uses an ideal gas with polytropic index $\gamma = \frac{5}{4}$ and initial high material density $\rho_0 = 19.32\ \text{g/cm}^3$[cite: 477, 480].

Specific Internal Energy: [cite: 481]

$$e(T,\rho) = 3.4\times 10^{13}\left(\frac{T}{\text{HeV}}\right)^{1.6}\left(\frac{\rho}{\text{g/cm}^3}\right)^{-0.14}\frac{\text{erg}}{\text{g}}$$

Rosseland Opacity Coefficient: [cite: 481]

$$\kappa_R(T,\rho) = 7200\left(\frac{T}{\text{HeV}}\right)^{-1.5}\left(\frac{\rho}{\text{g/cm}^3}\right)^{0.2}$$

Boundary Conditions & Self-Similar Evolution ($\tau = 0$)

Surface drive is maintained via a fixed thermal profile $T_s(t) = 1$ HeV[cite: 479, 485]. The dynamic radiation bath drive and the calculated mass front position trace as[cite: 488, 489]:

$$T_{\text{bath}}(t) = \left[1 + 0.169141\left(\frac{t}{\text{ns}}\right)^{-\frac{79}{192}}\right]^{1/4} T_s(t)$$

Simulation Setup: Case 2 ($\tau \approx 0.123$)

Time-Independent Surface Flux Environment

To simulate a completely steady, time-independent surface radiation energy flux ($c_S = 0$), the temporal power component is optimized analytically[cite: 490, 494]:

$$\tau = \frac{2 - 3\mu}{8 + 2\alpha + 2\beta + 3\beta\lambda - 3(4+\alpha)\mu} = \frac{158}{1285} \approx 0.12296$$

- **Material Invariance:** Identical baseline properties as Case 1 ($\gamma = \frac{5}{4}$, same e, same κ_R)[cite: 493].
- **Time-Dependent Surface Drive:** Scales upward explicitly as $T_s(t) = \left(\frac{t}{\text{ns}}\right)^{\frac{158}{1285}}$ HeV[cite: 496].

Self-Similar Validation Metrics

Analytical tracking metrics for the flux-matched drive are implemented as[cite: 496, 497]:

$$T_{\text{bath}}(t) = \left[1 + 0.204154\left(\frac{t}{\text{ns}}\right)^{-\frac{632}{1285}}\right]^{1/4} T_s(t)$$

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Summary

- **Complete Solution Matrix:** Provides a rigorous, closed-form analytic description of a subsonic radiation heat wave for generalized power-law boundaries[cite: 470].
- **Robust Decoupled Framework:** Separates the localized ablation and shock regions successfully, utilizing cross-boundary patching values[cite: 471].
- **Practical Utility:** Enables higher fidelity planning for laser-plasma and high-energy-density experiments while revealing the physical structure of ablation-driven shock waves[cite: 472].

Primary Literature Source

Full mathematical derivations and comprehensive simulation arrays are published in:

Phys. Plasmas 22, 082109 (2015) [cite: 473]